Three-Point Estimates — Worked Solution

Step 1 — Expected Duration (t■) per Activity

Formula: t■ = (O + M + P) / 3

Triangular distribution simple average. ★ marks critical-path activities whose t■ flows directly into the project expected duration.

★ SD1: Requirements Gathering

Formula: t■ = (O + M + P) / 3
Substitution: = (2.0 + 3.0 + 5.0) / 3
Result: t■ = 3.3333

★ SD2: Architecture Design

Formula: t■ = (O + M + P) / 3
Substitution: = (2.0 + 4.0 + 7.0) / 3
Result: t■ = 4.3333

SD3: Database Design

Formula: t■ = (O + M + P) / 3
Substitution: = (1.0 + 2.0 + 4.0) / 3
Result: t■ = 2.3333

★ SD4: Backend Development

Formula: t■ = (O + M + P) / 3
Substitution: = (5.0 + 8.0 + 14.0) / 3
Result: t■ = 9.0

SD5: Frontend Development

Formula: t■ = (O + M + P) / 3
Substitution: = (4.0 + 7.0 + 11.0) / 3
Result: t■ = 7.3333

SD6: API Integration

Formula: t■ = (O + M + P) / 3
Substitution: = (2.0 + 3.0 + 6.0) / 3
Result: t■ = 3.6667

★ SD7: Unit Testing

Formula: t■ = (O + M + P) / 3
Substitution: = (3.0 + 5.0 + 9.0) / 3
Result: t■ = 5.6667

★ SD8: Integration Testing

Formula: t■ = (O + M + P) / 3
Substitution: = (2.0 + 3.0 + 6.0) / 3
Result: t■ = 3.6667

★ SD9: Security Audit

Formula: t■ = (O + M + P) / 3
Substitution: = (1.0 + 2.0 + 4.0) / 3
Result: t■ = 2.3333

★ SD10: User Acceptance Testing

Formula: t■ = (O + M + P) / 3
Substitution: = (2.0 + 3.0 + 5.0) / 3
Result: t■ = 3.3333

SD11: Deployment & DevOps

Formula: t■ = (O + M + P) / 3
Substitution: = (2.0 + 3.0 + 4.0) / 3

Result: t■ = 3.0

★ SD12: Documentation & Training

Formula: t■ = (O + M + P) / 3

Substitution: = (2.0 + 3.0 + 5.0) / 3

Result: t■ = 3.3333

Step 2 — Variance (σ^2) per Critical-Path Activity

Formula: $\sigma^2 = (O^2 + M^2 + P^2 - OM - OP - MP) / 18$

Only critical-path activities contribute to project-level uncertainty. A large σ^2 signals a risky activity whose duration is hard to predict.

SD1: Requirements Gathering

Formula: $\sigma^2 = (O^2 + M^2 + P^2 - OM - OP - MP) / 18$

Substitution: $= (2.0^2 + 3.0^2 + 5.0^2 - 2.0\times3.0 - 2.0\times5.0 - 3.0\times5.0) / 18$

Result: $\sigma^2 = 0.3889$

SD2: Architecture Design

Formula: $\sigma^2 = (O^2 + M^2 + P^2 - OM - OP - MP) / 18$

Substitution: $= (2.0^2 + 4.0^2 + 7.0^2 - 2.0\times4.0 - 2.0\times7.0 - 4.0\times7.0) / 18$

Result: $\sigma^2 = 1.0556$

SD4: Backend Development

Formula: $\sigma^2 = (O^2 + M^2 + P^2 - OM - OP - MP) / 18$

Substitution: $= (5.0^2 + 8.0^2 + 14.0^2 - 5.0\times8.0 - 5.0\times14.0 - 8.0\times14.0) / 18$

Result: $\sigma^2 = 3.5$

SD7: Unit Testing

Formula: $\sigma^2 = (O^2 + M^2 + P^2 - OM - OP - MP) / 18$

Substitution: $= (3.0^2 + 5.0^2 + 9.0^2 - 3.0\times5.0 - 3.0\times9.0 - 5.0\times9.0) / 18$

Result: $\sigma^2 = 1.5556$

SD8: Integration Testing

Formula: $\sigma^2 = (O^2 + M^2 + P^2 - OM - OP - MP) / 18$

Substitution: $= (2.0^2 + 3.0^2 + 6.0^2 - 2.0\times3.0 - 2.0\times6.0 - 3.0\times6.0) / 18$

Result: $\sigma^2 = 0.7222$

SD9: Security Audit

Formula: $\sigma^2 = (O^2 + M^2 + P^2 - OM - OP - MP) / 18$

Substitution: $= (1.0^2 + 2.0^2 + 4.0^2 - 1.0\times2.0 - 1.0\times4.0 - 2.0\times4.0) / 18$

Result: $\sigma^2 = 0.3889$

SD10: User Acceptance Testing

Formula: $\sigma^2 = (O^2 + M^2 + P^2 - OM - OP - MP) / 18$

Substitution: $= (2.0^2 + 3.0^2 + 5.0^2 - 2.0\times3.0 - 2.0\times5.0 - 3.0\times5.0) / 18$

Result: $\sigma^2 = 0.3889$

SD12: Documentation & Training

Formula: $\sigma^2 = (O^2 + M^2 + P^2 - OM - OP - MP) / 18$

Substitution: $= (2.0^2 + 3.0^2 + 5.0^2 - 2.0\times3.0 - 2.0\times5.0 - 3.0\times5.0) / 18$

Result: $\sigma^2 = 0.3889$

Step 3 — Project Variance (σ^2_project)

Formula: $\sigma^2\_project = \Sigma\, \sigma^2(\text{critical-path activities})$

Substitution: $= 0.3889 + 1.0556 + 3.5 + 1.5556 + 0.7222 + 0.3889 + 0.3889 + 0.3889$

Result: $= 8.389$

Project variance is the sum of variances along the critical path. Parallel non-critical activities do not add uncertainty to project duration.

Step 4 — Project Standard Deviation (σ)

Formula: $\sigma = \sqrt{(\sigma^2\_project)}$

Substitution: $= \sqrt{(8.389)}$

Result: = 2.8964

Standard deviation in the same units as duration. Rule of thumb: the project will finish within ±1σ of the expected duration about 68% o

Step 5 — Project Expected Duration

Formula: T_project = Σ t■(critical-path activities)

Substitution: = 3.3333 + 4.3333 + 9.0 + 5.6667 + 3.6667 + 2.3333 + 3.3333 + 3.3333

Result: = 34.9999

The project's expected completion time, based on the sum of expected durations along the critical path.